Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/24/2013
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Project Title:	Effects of Space Radiation on the Function of Human Brain Capillaries and the Blood-brain-barrier: Implications for Neurodegenerative Disorders		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ponnaiya, Brian (Columbia University) Smilenov, Lubomir (Columbia University)		
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Task Description:	This proposal is aimed at determining the effects of space radiation on the function of human brain blood vessels (capillaries) and the Blood-Brain-Barrier (BBB). The brain vascular system is crucial to healthy functioning of the brain and its dysfunction is not only a primary event in a range of neurodegenerative diseases but also an important influencing factor in many others. Furthermore, vessel pathologies involve common susceptibility factors such as oxidative stress and inflammation so any radiation effects on capillaries can be influenced by other preexisting conditions. Here, we propose to investigate the effects of space radiation on angiogenesis and BBB function using human brain blood vessel cells in 2-dimensional and 3-dimensional tissue models. Since the brain vascular system is essentially an integrated multi-cell system we also propose to utilize other human brain cell types in tissue models to create the most realistic human vessel model possible. We have focused on the main aspects of vessel function that are relevant to the research emphases of acute and late risks to the brain. That is, functioning of the BBB, blood vessel development and the compound effects of preexisting conditions such as inflammation. A correctly functioning blood system supplies the brain and other tissues with oxygen and nutrients while preventing deleterious blood-borne molecules from entering the brain and causing possible damage to brain cells. Tight junctions between the blood vessel cells are formed by proteins that make a tight seal and prevent leakiness. We will determine the effect of space radiation on blood vessels of the brain and its possible impact on neurodegeneration by testing for vessel formation (angiogenesis) vessel leakiness, and the influence of other factors such as inflammation molecules after exposure to both galactic cosmic radiation and radiation from the sun.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gamma radiation studies Studies to optimize assay conditions using gamma radiation revealed a previously unknown effect of gamma photons on human blood vessels. It is the first demonstration of a transient and rapid effect of gamma radiation on human endothelial barriers that involves cell detachment and loss of the cell adhesion molecule - PECAM-1. This effect occurred at the fractionated radiotherapy dose of 2 Gy and therefore has the potential to be of clinically relevant importance considering the role of PECAM-1 in many of the cellular processes involved in pathologies seen at longer times after radiation therapy. As such it could be a target for treatments that alleviate these pathologies. A second potential benefit is that of drug delivery to tumors. A temporary opening of the endothelial barrier by the dual effects of low temperatures and gamma radiation could be a means by which impermeable chemotherapy drugs could be delivered directly to tumors. Tissue models The development of 3-Dimensional human tissue models from normal human cells and stem cells has great potential in many fields of medical research. Tissue models can more accurately depict human tissue since the cells can be arranged spatially as they would be in vivo and can interact with each other as they would in the human body. A neurovascular unit can be used for basic research on many aspects of the human brain. These include regeneration, synaptic function, and degeneration. Because the tissue model is derived from individual cells, each cell type can be altered genetically before it is incorporated into the model.
Task Progress:	Endothelial barrier studies Gamma radiation While waiting for beam time we tested the assays using Gamma radiation. This was intended for several reasons. It allowed us to allow us to determine how the endpoints relate to each other with regard to sensitivities. We were also able to streamline the assays before going to NSRL. In addition, it gave us a counterpoint with which to compare the effects of ion particles. Finally, our investigations of the early effects of photons revealed the first demonstration of a transient and rapid effect of gamma radiation on human endothelial barriers that involves cell detachment and loss of the cell adhesion protein - PECAM-1 (or CD31). In the last year we have completed these studies by identifying loss of PECAM-1 in 3-D human vessel models. Such an effect has the potential to be clinically relevant to longer-term effects of radiotherapy. This work has been published in the journal – Microvascular Research. Ion particles The potential clinical importance of the effect of gamma radiation on PECAM-1 in human vessels prompted us to investigate whether other photons or charged particles would also cause uncoupling of PECAM-1 or any other acute effects. We have assessed the effects of low LET radiation on human endothelial barrier function in two extremes of physiological conditions. Trans endothelial electrical resistance measurements (TEER) measurements showed that photons but not charged particles (protons and He ions) induce a transient drop in TEER 3 hours after exposure regardless of the cell environmental conditions. This drop in TEER indicates the uncoupling of PECAM-1 and a compromise of the endothelial barrier. Both charged particles and photons induced an under-damped oscillation in TEER in the presence of serum and growth factors but not in serum-free conditions. The work in this section is currently being prepared as a manuscript. Fe ions were also examined for their effect on endothelial barrier function. A dose of 75 cGy was sufficient to affect TEER, permeability to marker molecules and relocalization of tight junctions. With a dose of 25 cGy however, only permeability to 3 kDa marker molecules and tight junction protein relocalization was observed. No changes were seen with a dose of 10 cGy. The threshold dose for deleterious effects on the human endothelial barrier is considered to be 25 cGy. This work is currently being prepared as a manuscript. Angiogenesis studies We have previously shown that low LET protons and high LET Fe ions inhibit the formation of brain capillaries by different mechanisms. This work has been written into a manuscript and submitted to the Journal – Vascular Cell. Ongoing studies are aimed at determining the LET ranges for each type of inhibition, To date low LET charged particles in the range 0.2 to 0.89 (protons 1 GeV and 200 MeV, Helium 1 GeV) inhibit vasculogenesis by inhibiting motile tip activity but not tubulogenesis. Higher LET particles in the range 15-150 (1 GeV Fe ions, 1 GeV Si ions, 250

	MeV and 1 GeV O ions) by inhibiting tubulogenesis activity but not motile tip activity. 100 MeV He ions (LET = 2.9) did not inhibit either, vasculogenesis was not inhibited by a dose of 1 Gy.
Bibliography Type:	Description: (Last Updated: 03/04/2024)
Abstracts for Journals and Proceedings	Sharma P, Templin T, Grabham P. "Acute effects of low LET radiation on human endothelial barrier function." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Sharma P, Templin T, Grabham P. "Acute effects of ionizing radiation on human endothelial barrier function." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Articles in Peer-reviewed Journals	Sharma P, Templin T, Grabham P. "Short term effects of gamma radiation on endothelial barrier function: uncoupling of PECAM-1." Microvascular Research. 2013 Mar;86:11-20. Epub 2012 Dec 5. http://dx.doi.org/10.1016/j.mvr.2012.11.007 ; PubMed PMID: 23220351 , Mar-2013